

Junhao (Leon) Han

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SUMMARY

Software developer and Master's student in Computing at Queen's University with experience building backend systems, web applications, and AI-integrated software. Developed a PKP-approved open-source OJS plugin using PHP and JavaScript to improve the academic paper review process. Built hackathon and research projects using Node.js, REST APIs, WebSockets, LLM integration, and PyTorch. Always interested in learning new technologies and applying them to practical software projects.

TECHNICAL SKILLS

Languages: Python, Java, PHP, JavaScript, C#, SQL, HTML/CSS
Web Development: React, Node.js, Vite, REST APIs, WebSockets, OJS
AI/ML: PyTorch, Neural Networks, LLM Integration, RAG, Embeddings, Model Training, Data Analysis
Tools & Platforms: Git, Docker, Linux, AWS, VS Code, Visual Studio, IntelliJ
Additional: English and Mandarin Chinese; golf and tennis

EXPERIENCE

- Backend Web Developer** May 2026 – Present
Encounters in Theory and History of Education Journal, Queen's University Kingston, ON
- Designed and developed a PKP-approved OJS 3.3 plugin using PHP and JavaScript, enabling editors and reviewers to preview static web projects directly within the journal's editorial workflow.
 - Built PHP routes for preview rendering and static asset delivery, resolving nested `index.html` entry points and serving HTML, CSS, JavaScript, images, fonts, audio, and video.
 - Support ongoing OJS UI updates using HTML, CSS, and JavaScript to improve usability across editorial workflows.
- Teaching Assistant** Sept. 2025 – May 2026
Queen's University Kingston, ON
- CISC 801 – AI and Machine Learning: Guided graduate students in intelligent agents, prompt engineering, embeddings, vector search, RAG pipelines, and LLM application development and debugging.
 - CISC 365 – Algorithms I: Supported grading and explained divide-and-conquer, greedy algorithms, dynamic programming, and complexity analysis.
- Investment Associate Intern, Tech Department** May 2025 – Aug. 2025
Sixty Degree Capital Toronto, ON
- Evaluated IT startups by analyzing software products, technology stacks, and technical feasibility, and discussed product architecture and development plans with startup CTOs.
 - Researched and delivered a 30-minute presentation on a stablecoin startup, translating its blockchain infrastructure, business model, and regulatory risks into clear findings for the investment team.

PROJECTS

- Semantic Vehicular Communication under Intelligent Jamming** | *Python, PyTorch, Optimization* 2026
- Conducted research for a paper on secure semantic vehicular communication and implemented a three-phase adversarial learning pipeline in Python, improving transmission accuracy by up to **17.6%** over fixed baselines.
 - Trained a residual neural network for vehicle re-identification, computed feature-importance scores using gradient-based chain-rule analysis, and modeled jammer retraining as a Markov decision process under bandwidth constraints.
- MindBrief** | *JavaScript, Node.js, ElevenLabs API* 2026
- Built a web application for structured group brainstorming for KingHacks 2026 using the *Six Thinking Hats* method.
 - Implemented a Node.js WebSocket server that streamed browser microphone audio to the ElevenLabs real-time speech-to-text API and generated LLM-based meeting summaries from the resulting transcripts.
 - Developed participant setup, drag-and-drop role assignment, and guided speaking workflows in the front end.

EDUCATION

- Queen's University** Kingston, ON
Master of Science in Computing Sept. 2025 – Dec. 2026 (Expected)
- Queen's University** Kingston, ON
Bachelor of Computing (Honours) Sept. 2021 – May 2025